

Insulin and Glycemic Management

Policy Number	POL-PH-031	Owner / Department	Pharmacy / Nursing
Effective Date	2023-07-10	Version	2.3
Last Revised	2026-03-18	Next Review	2028-03-18
Approved By	Pharmacy & Therapeutics Committee	Applies To	All prescribers, pharmacists, and nurses managing insulin, oral hypoglycemics, or inpatient glucose in any care area

1. Purpose

To standardize safe prescribing, dispensing, administration, and monitoring of insulin and other glucose-lowering agents, and to define the response to hypoglycemia. Insulin is a high-alert medication on the ISMP acute-care list.

2. Definitions

- **Hypoglycemia** — blood glucose below 70 mg/dL; severe hypoglycemia is below 40 mg/dL or any value requiring assistance.
- **Basal-bolus regimen** — scheduled long-acting (basal) insulin plus mealtime (bolus/nutritional) and correction insulin.
- **Concentrated insulin** — products more concentrated than U-100 (e.g., U-200, U-300, U-500) that require special handling.
- **Point-of-care (POC) glucose** — a bedside capillary glucose measurement used to guide dosing.

3. Policy and Procedure

Insulin orders must specify product, concentration, dose in units, and timing relative to meals; the abbreviation “U” for units is prohibited — write “units” in full. Insulin is dispensed in ready-to-use strengths; U-500 and other concentrated insulins are stored separately from U-100.

1. Confirm a current POC or lab glucose before administering mealtime or correction insulin.
2. Perform an independent double check of product, concentration, and units for every IV insulin infusion and for all pediatric doses.
3. Match nutritional insulin to actual intake; hold or adjust the nutritional dose if the patient is NPO or not eating.
4. Use only insulin-specific (unit-marked) syringes or the approved pen device; never draw insulin into a tuberculin or standard syringe.
5. Monitor glucose per the ordered schedule and the hypoglycemia protocol.

IMPORTANT — For blood glucose below 70 mg/dL, treat immediately per the nurse-driven hypoglycemia protocol (15 g fast-acting carbohydrate if the patient can swallow safely, or IV dextrose / IM glucagon if not), recheck glucose in 15 minutes, and notify the provider. Do not delay treatment to obtain an order.

4. Glucose Monitoring and Response Parameters

POC glucose (mg/dL)	Action
Less than 54	Severe hypoglycemia — treat STAT, give IV dextrose/glucagon, notify provider, recheck q15 min.
54–69	Hypoglycemia — 15 g oral carbohydrate, recheck in 15 minutes, repeat until above 70.
70–180	Target inpatient range — continue ordered regimen.
Greater than 300	Marked hyperglycemia — confirm value, assess for DKA/HHS, notify provider.

5. Severity & Reporting

Classify glycemic and insulin events on the POL-RM-013 scale and grade medication errors by the NCC MERP index per POL-PS-003.

Severity	Example	Notify / timeframe
SEV-1	Severe hypoglycemia causing seizure, coma, or death; 10-fold insulin overdose.	Provider, Nursing Supervisor & Risk Mgmt within 1 hour.
SEV-2	Hypoglycemia requiring IV dextrose/glucagon or prolonging stay.	Provider & Risk Mgmt within 4 hours.
SEV-3 / Near Miss	Delayed glucose check, no harm; wrong insulin intercepted before administration.	Event report within 24 hours.

6. Documentation Requirements

- Insulin product, concentration, dose in units, route, and timing relative to meals.
- POC/lab glucose value and time supporting each dose.
- Independent double check for IV insulin and pediatric doses, with verifier names.
- Hypoglycemia treatment given, recheck values, and provider notification.

7. References

- ISMP List of High-Alert Medications in Acute Care Settings.
- ISMP List of Error-Prone Abbreviations, Symbols, and Dose Designations (“U” for units).
- The Joint Commission NPSG.03.06.01 (medication reconciliation) and medication-management standards.
- Related: POL-RM-013 (severity & reporting); POL-PS-003 (medication-error reporting); POL-PH-032 (high-alert medications).